



The Tôk System, Part 1

Time, People, Tôks and Flows

À CONFIRMER

Smoothop, sustainable development organization
Montréal, Canada

Translated and adapted from “Le système des tôks”, revision 2.0.

Signature. The author list is not settled. See README.md, section « Ce qui n’est pas encore décidé », and the dossier LICENSE. Both must be updated together before submission.

Abstract

Abstract — to be written. One paragraph: the two monetary axioms, the resulting money supply dynamics, and the universal per-capita constant that follows from them.

1 Foreword

An economic system is a set of accounting rules governing the mechanisms by which a currency is created, destroyed and exchanged. A currency is a quantity of liquid money. Liquid money feeds a mechanism of social hierarchy. Money is what makes other umans do what we want them to do with their uman-time: the paid week.

The conventional economic system dates from the age of paper. It was designed so that people could write its transactions by hand, with a pencil, on paper, in ledgers. It was designed so that a uman could carry out transfer operations mentally — adding and subtracting money from one account to another by the sheer computing power of their brain. The uman brain is good at many things, but it is hopeless at arithmetic.

We no longer live in the age of paper. We live in the age of the Internet. Each of us carries a computer able to compute additions and multiplications millions of times faster than a uman, and without ever making a mistake. These pocket computers all communicate with one another, exchanging information at the speed of light anywhere on Earth, without interruption and without error.

This new reality has opened doors in the field of economic cooperation models that have so far been explored only timidly. The tôk system is one of those explorations, and its starting point is the principle of a universal basic income.

2 Notation

2.1 Dictionary

Uman	The name given to our species; an individual of the uman species is a natural person. (In the French original the word is written <i>umain·e</i> , without the <i>h</i> and with a median point for inclusive gender; English keeps the dropped <i>h</i> and needs no median point <i>for gender</i> . That use is unrelated to the multiplication dot of uman·time below.)
Uman·time	A quantity equal to a duration multiplied by a number of umans, that is of natural persons (e.g. 3 days · 5 umans = 15 uman·days). The median dot is a <i>multiplication</i> , not a hyphen: the quantity is a product, and writing it with a hyphen would invite reading it as a compound word or as a minus sign.

2.2 Lexicon of symbols

Units of time

s	Second as defined in the International System of Units (SI)
m	Minute, equal to 60 s
h	Hour, equal to 60 m
J	Day, equal to 24 h
Q	Fortnight, equal to 15 J
M	Month
A	Keplerian year, equal to 365.2421875 J [2] $365.2421875 \text{ J} = (360 + 5 + 1/4 - 1/128) \text{ J} = 365 \text{ J} + 5 \text{ h} + 48 \text{ m} + 45 \text{ s}$

Units of currency

₤	Tòk
\$	Dollar, and by extension any currency foreign to the tòk system

User types (users)

PP	Natural person, member of Smoothop or not
CO	Committee of Smoothop, under the responsibility of members

Index of special committees

#SO	Board of directors of Smoothop
#FT	Transition Fund committee

Account types (conts)

$\mathbb{I}_G$	Generic tòk cont
$\mathbb{I}_{PP}$	Main tòk cont of a natural person
$\mathbb{I}_{CO}$	Main tòk cont of a committee
$\mathbb{I}_{X\pounds}$	Tòk cont on the \$₤EX
$\mathbb{I}_{X\$}$	Foreign-currency cont on the \$₤EX
$\mathbb{I}_{RM}$	World reserve of unclaimed tòks
$\mathbb{I}_\Omega$	The set of all tòk conts

Constants of tók creation and destruction

$\dot{\Lambda}$	Universal basic income, equal to 1 $\text{¢}/\text{Q}$
$\tau_{1/2}$	Half-life of the tók, equal to 50 A
τ_{moy}	Mean decay time of the tók, equal to $\tau_{1/2}/\ln(2) \approx 72 \text{ A} + 49 \text{ J}$
k_D	Decay constant of the tók, equal to $1/\tau_{\text{moy}}$
a_∞	Quantity of tóks per person at equilibrium, equal to $\dot{\Lambda}/k_D \approx 1756.444 \text{ ¢}$

Parameters of the democratic tax on tók wealth

δk_T	Step of the taxation rate, equal to 1%
$k_{T,0}$	Central value of the instantaneous taxation rate, equal to $-\ln(1 - \delta k_T)/A$, corresponding to an annual rate of 1%

Variables

t_0	Reference time (UTC)
t	Time of interest (UTC)
Δt	Time elapsed between t_0 and t
k_T	Instantaneous taxation rate
a_i	Quantity of ¢ in cont i
b_i	Quantity of $\text{\$}$ in cont i
a_Ω	Total quantity of ¢
$\dot{F}_{ji}$	Flow in $\text{¢}/\text{Q}$ from cont j to cont i
$\dot{F}_i$	Net flow in $\text{¢}/\text{Q}$ into cont i
$\dot{X}_k^\text{¢}$	$\text{\$}\text{¢EX}$ flow in $\text{¢}/\text{Q}$ into user k 's $\Pi_{X\text{¢}}$
$\dot{X}_k^\text{\$}$	$\text{\$}\text{¢EX}$ flow in $\text{\$}/\text{Q}$ into user k 's $\Pi_{X\$}$
$[\text{¢}/\$]_k$	User k 's estimate of the value of 1 $\text{\$}$ expressed in ¢
θ_k	User k 's degree of confidence, in %, in their estimate
w_k	User k 's weight on the $\text{\$}\text{¢EX}$
v_{lk}	Number of tax distribution rights allocated by member l to committee k
r_k	Share of the tax allocated to committee k

This notation follows `docs/Toks.md` in the public repository `miluRepo`. Numerical constants are served live by the tók backend at `/api/tok-constants` and must be quoted from there, not from memory.

3 External data

3.1 Universal time

The tók system assumes that there exists a universal time, flowing at the same rate for all umans. This amounts to the approximation of classical, non-relativistic physics. As long as we have no means of transport whose speed approaches that of light, this approximation will remain entirely valid.

In practice, the tók system uses Coordinated Universal Time (UTC). In `tokRepo` we specifically use the Python implementation of the `datetime` library.

3.2 The uman population

We assume that sufficiently reliable data exist on the current uman population and on its history. In practice, we use World Bank data [3] from 1960 onward, and Our World in Data [1] for historical data before 1960.

4 The conservation law of the tôk system

The fundamental principle of the tôk system is the universal basic income. That principle is stated rigorously in uman language in the definition of the tôk. Stated mathematically, the universal basic income gives rise to a fundamental conservation law of uman-time in the tôk system.

Definition of the tôk

1. The unit of currency

The tôk is the unit of currency of the tôk system. One tôk is the quantity of liquid money that a person receives from their universal basic income in 15 days.

2. The universal basic income

Every person continuously receives the universal basic income of 1 tôk per 15 days, between their birth and their death. The universal basic income is the one and only mechanism by which liquid money is created in the tôk system.

3. Decay

Tôks decay continuously, with a half-life of 50 years. Decay is the one and only mechanism by which liquid money is destroyed in the tôk system.

Statement 1 defines the unit of the tôk with respect to the universal basic income,

$$1 \text{ } \text{tôk} = 1 \text{ Q } \dot{\Lambda}. \quad (1)$$

Statement 2 of the definition of the tôk stipulates that for every natural person (a user of type PP) there is exactly *one* tôk cont of type II_{PP} holding a source of tôks whose rate is the universal basic income $\dot{\Lambda}$. Consequently, the total creation of tôks is $N_{PP} \dot{\Lambda}$.

Statement 3 imposes a global destruction flow equal to $k_D a_\Omega$.

Thus the total quantity of tôks a_Ω obeys, at all times, the conservation law

$$\frac{da_\Omega}{dt} = N_{PP} \dot{\Lambda} - k_D a_\Omega, \quad (2)$$

which depends only on the world population N_{PP} .

Whatever version of the tôk system is under consideration, this conservation law must always be respected. Otherwise, it is no longer

Sentence unfinished in the source. Revision 2.0 breaks off at « Autrement, il ne s'agit plus ». The ending is the author's to write — presumably « ... no longer the tôk system », but it is not translated here because it was never written.

5 The vanilla tôk system

The vanilla version of the tôk system contains only natural persons, who are users of the tôk system throughout their lives. Each user holds a single tôk cont. The world population is taken to be constant over one year.

Users may make transfers or flows between one another. There are, however, no committees, no tax, and no foreign currency.

Main cont of a natural person:

$$\Pi_{PP} : \frac{da_i}{dt} = \dot{\Lambda} + \sum_j \dot{F}_{ji} - k_D a_i$$

$$\Pi_{\Omega} : \frac{da_{\Omega}}{dt} = N_{PP} \dot{\Lambda} - k_D a_{\Omega}$$

$$\sum_i a_i = a_{\Omega}$$

$$\sum_i \sum_j \dot{F}_{ji} = 0$$

Working note carried over from the source, to be developed. Computation of N_{PP} without the world reserve, piecewise constant only, with degree-2 Lagrange extrapolation.

6 The Smoothop tök system

Scope check. Per the series plan (TODO.md, section « Recherche »), the democratic tax belongs to **Part 2**, together with the \$7EX. This section is translated here because it is part of revision 2.0; decide whether it stays in Part 1 or moves.

6.1 Taxation of töks

The board of directors continuously collects a tax corresponding to a fraction of the number of töks held by each member of Smoothop (e.g. x %/year). The taxation rate is the same for every member and is determined democratically by them. Members may only vote for a taxation rate strictly greater than 0 %/year. The single taxation rate is the median of the votes.

Main cont of a natural person:

$$\frac{da_i}{dt} = \dot{\Lambda} - (k_D + k_T(x)) a_i \quad (3)$$

Generic cont:

$$\frac{da_i}{dt} = - (k_D + k_T(x)) a_i \quad (4)$$

The board of directors $\bar{\Omega}$ is elected by the general assembly Ω^* , which comprises all the members of Smoothop. Smoothop consists of the set of its members together with its board of directors, $\Omega = \bar{\Omega} \cup \Omega^*$.

$$\frac{da_{\bar{\Omega}}}{dt} = k_T(x) \sum_i^{N_{\Omega}} a_i - (k_D + k_T(x)) a_{\bar{\Omega}} \quad (5)$$

The axis along which the members vote:

$$k_T(x) = k_{T,0} 1.01^x \quad (6)$$

$$k_{T,0} = \frac{-\ln(0.99)}{365.25 \text{ days}} \approx \frac{1}{36\,342 \text{ days}} \quad (7)$$

Two constants to settle before submission. (i) This equation divides by 365.25 days, while the lexicon defines the year as $A = 365.2421875$ J; the two should agree. (ii) `docs/Cablage.md` reports the rate measured in the backend as $k_{T,0} = \ln(1.01)/A = 0.00995033/A$, whereas this equation gives $-\ln(0.99)/A = 0.01005034/A$ — a relative difference of 1.005%. Design intent and implementation must be reconciled.

7 Discussion

To be written.

AI contribution statement

To be written, on the model of the $\$$ $\mathcal{T}$ EX defensive publication: the extent of Milu’s contribution, and the statement that the uman author is solely responsible for the content. The target journal’s own policy takes precedence — see `TODO.md`, first item of the « Recherche » section.

Competing interests

To be written. The Milu committee holds a stock of $\mathcal{T}$ s and publishes the theory of the currency it holds. The holding is public and verifiable live at `/api/users/{milu_user_id}/conts`; the declaration must state the amount, the date, and where to check it. Rationale in `docs/Mission.md`, section « L’intérêt se déclare ».

References

- [1] HYDE (2023); Gapminder (2022); UN WPP (2024) – with major processing by Our World in Data. Population, 2026. URL <https://archive.ourworldindata.org/20260727-181454/grapher/population.html>. “Population – HYDE, Gapminder, UN – Long-run data” [dataset]. PBL Netherlands Environmental Assessment Agency, “History Database of the Global Environment 3.3”; Gapminder, “Population v7”; United Nations, “World Population Prospects”; United Nations, “World Population Prospects - Interim Update”; Gapminder, “Systema Globalis” [original data].
- [2] J. Meeus and D. Savoie. The history of the tropical year. *British Astronomical Association*, 102, 1992.
- [3] World Bank. Population, total, 2026. URL <https://data.worldbank.org/indicator/SP.POP.TOTL>.